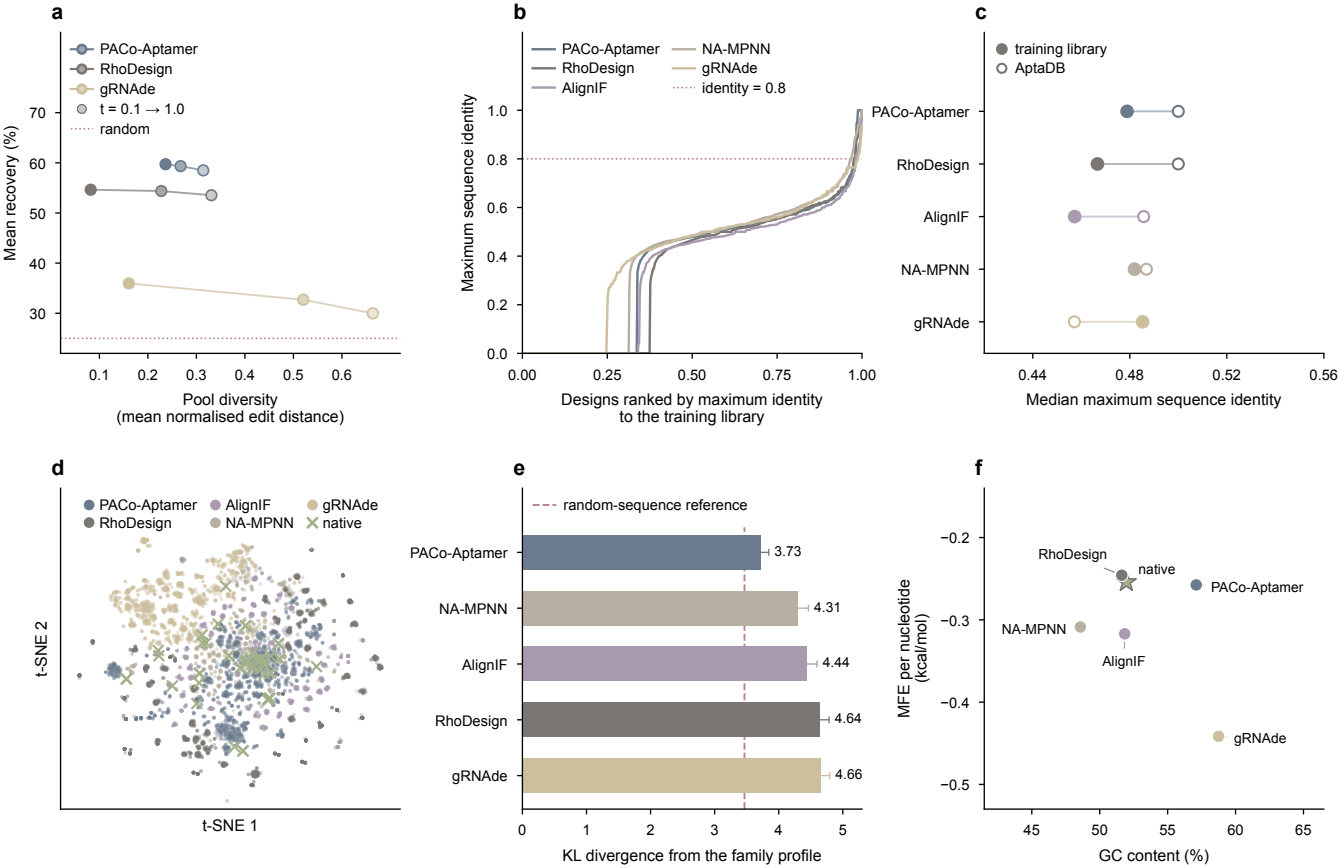

a
PACo-Aptamer
RhoDesign
gRNAde
t = 0.1 → 1.0
random
Mean recovery (%)
70
60
50
40
30
0.1
0.2
0.3
0.4
0.5
0.6
Pool diversity
(mean normalised edit distance)
b
PACo-Aptamer
RhoDesign
AlignIF
NA-MPNN
gRNAde
identity = 0.8
Maximum sequence identity
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.25
0.50
0.75
1.00
Designs ranked by maximum identity
to the training library
c
training library
AptaDB
PACo-Aptamer
RhoDesign
AlignIF
NA-MPNN
gRNAde
0.44
0.48
0.52
0.56
Median maximum sequence identity
d
PACo-Aptamer
RhoDesign
AlignIF
NA-MPNN
gRNAde
native
t-SNE 2
t-SNE 1
e
random-sequence reference
PACo-Aptamer
NA-MPNN
AlignIF
RhoDesign
gRNAde
3.73
4.31
4.44
4.64
4.66
0
1
2
3
4
5
KL divergence from the family profile
f
RhoDesign
native
PACo-Aptamer
NA-MPNN
AlignIF
gRNAde
MFE per nucleotide
(kcal/mol)
−0.2
−0.3
−0.4
−0.5
45
50
55
60
65
GC content (%)